INCH-POUND

MIL-PRF-55339/43B
w/AMENDMENT 1
24 January 2025
SUPERSEDING
MIL-PRF-55339/43B
28 January 2015

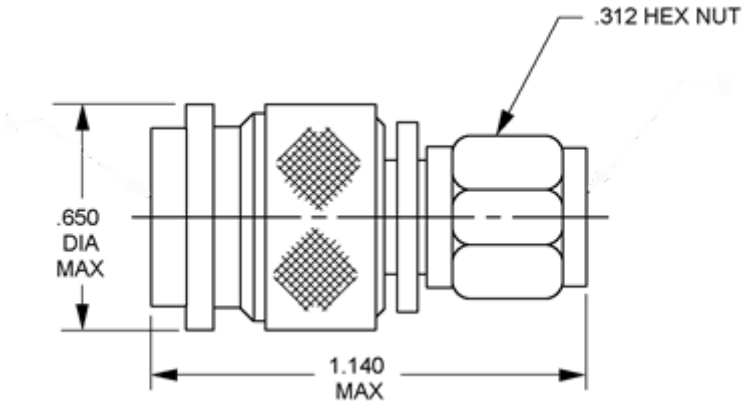
PERFORMANCE SPECIFICATION

ADAPTERS, CONNECTOR, COAXIAL, RADIO FREQUENCY, (BETWEEN SERIES SMA (MALE) TO SERIES TNC (MALE)), CLASS 2, STRAIGHT PLUG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55339.

INACTIVE FOR NEW DESIGN
AFTER
17 OCTOBER 1991



NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial representations are for reference purposes only.
3. Interface shall be in accordance with MIL-STD-348.
4. M55339/43–30001 and -50001: Three holes .016 minimum diameter, equally spaced, are required for safety wiring after mating. Location on coupling nut optional.

FIGURE 1. General configuration.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.
Frequency range: 0 to 11 GHz.
Voltage rating: 335 volts rms maximum working voltage at sea level, 85 volts rms at 70,000 feet.
Temperature range: -65° to +165°C.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Center contact retention.
Axial force – 6 pounds minimum.

Force to engage and disengage.
Longitudinal force – Not applicable.
Torque – 2 inch-pounds maximum.

Coupling proof torque.
Series SMA – 15 inch-pounds.
Series TNC – 15 inch-pounds.

Inspection conditions.
Coupling torque – Series SMA – 7 to 10 inch-pounds, Series TNC – 4 to 6 inch-pounds.

Mating characteristics.
Series TNC.
Center contact:
Test ring ID 319 maximum, 16 microinch finish.
Insertion force 5 pounds maximum when inserted a minimum of .093.
Contacts with slotted members shall contact a .324 minimum diameter ring within .031 of their tip ends.

Permeability: Not to exceed 2.

Insulation resistance: 5,000 megohms minimum.

Voltage standing wave ratio (VSWR): 1.15 + .015 F(GHz) at .5 to 11 GHz.

RF leakage: -60 dB minimum tested at a frequency of 2 to 3 GHz.

RF insertion loss: $.06 \sqrt{F\ (GHz)}$ dB maximum at 6 GHz.

Durability: 500 cycles minimum at 12 cycles per minute maximum.

Dielectric withstanding voltage: 1,000 volts rms, minimum.

Contact resistance In milliohms, maximum.

	Initial	After environment
Center contact	4.5 1/	6.0
Outer contact	2.2	--

1/ Two center contacts must be mated to the center conductor under test therefore doubling center contact resistance.

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Vibration, high frequency: MIL-STD-202-204, test condition D, interruptions – 1 µs, maximum.

Shock (specified pulse): MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition C.

Moisture resistance: 200 megohms, minimum.

Corona level:
Voltage – 375 volts minimum.
Altitude – 70,000 feet minimum.

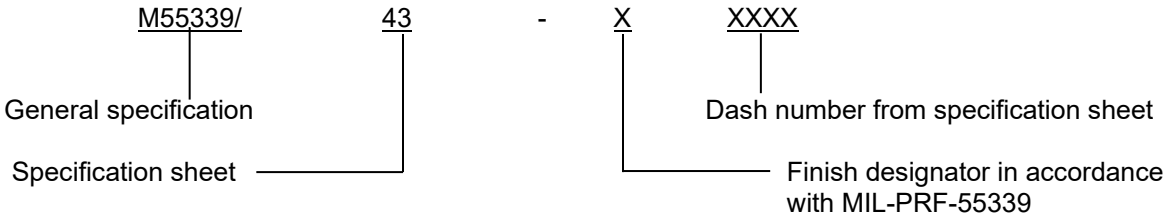
RF high potential withstanding voltage:
RF voltage – 1,000 volts rms minimum.
Frequency – 5 to 7.5 MHz.

Salt spray (corrosion): MIL-STD-202-101, test condition B.

Coupling mechanism retention force:
Series SMA – 60 pounds minimum.
Series TNC – 100 pounds minimum.

Part or Identifying Number (PIN): M55339/43-30001 (safety wire holes (SMA and TNC)).
-30101 (without safety wire holes)
M55339/43-50001 (safety wire holes (SMA and TNC)).
-50101 (without safety wire holes)

PIN configuration:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents: In addition to MIL-DTL-55339, this document references the following:

MIL-STD-202-101
MIL-STD-202-107
MIL-STD-202-204
MIL-STD-202-213
MIL-STD-348

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2025-008)

Review activities:
Army - AR, AT, EA, MI
Navy - AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

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